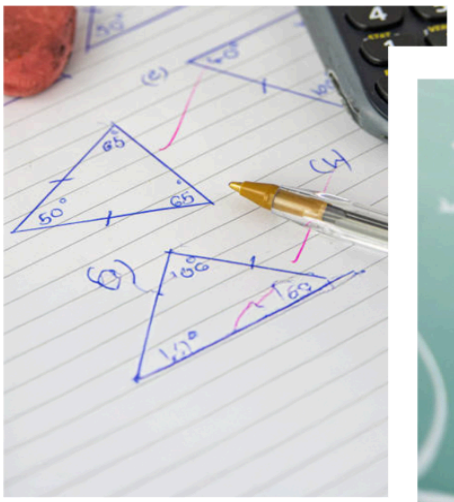
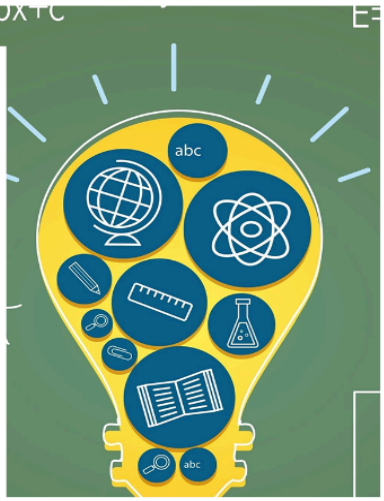
testbook PASS PRO MAX

Quantitative Aptitude

Formula – Guide

Mensuration 3D





Mensuration 3D

Cube

- $Total\ Surface\ Area\ =\ 6a^2$
- $Lateral\ Surface\ area\ of\ cube\ =\ 4a^2$
- $Volume\ =\ a^3$
- $Face\ Diagonal\ =\ a\sqrt{2}$
- $Space\ Diagonal\ =\ a\sqrt{3}$

Cuboid

- $Surface\ area\ =\ 2(lb\ +\ lh\ +\ bh)$
- $Volume\ =\ l \times b \times h$
- $Space\ Diagonal\ =\ \sqrt{l^2\ +\ b^2\ +\ h^2}$
- $Face\ Diagonal\ =\sqrt{l^2\ +\ b^2}$
- $(l + b + h)^2 = l^2 + b^2 + h^2 + 2\ (lb + bh + hl)$
- $(\text{sum of dimensions})^2 = (\text{Diagonal})^2 + \text{Total Surface Area}$

Cylinder

- $Lateral\ Surface\ Area\ =\ 2\pi rh$
- $Surface\ Area\ =\ 2\pi r(h\ +\ r)$
- $Volume\ =\ \pi r^2 h$

Hollow Cylinder

- $Lateral\ Surface\ Area\ =\ 2\pi rh\ +\ 2\pi Rh\ =\ 2\pi h\ (R\ +\ r)$
- $Surface\ Area\ =\ 2\pi h\ (R\ +\ r)\ +\ 2\pi(R^2 - r^2)$
- $Volume\ of\ hollow\ cylinder\ =\ \pi r^2 h$
- $Volume\ of\ material\ of\ hollow\ cylinder = \pi(R^2 - r^2)h$

Cone

- $Slant\ Height(l) = \sqrt{r^2 + h^2}$
- $Lateral\ Surface\ Area = \pi rl$
- $Total\ surface\ area = \pi r(r + l)$
- $Volume = \frac{1}{3}\pi r^2 h$

Frustum:

- $Lateral\ Surface\ Area = \pi(R + r)\ l$
- $Total\ surface\ area = \pi l(R + r) + \pi(R^2 - r^2)$
- $Volume = \frac{1}{3}\pi(R^2 + r^2 + Rr)\ h$
- $Slant\ height(l) = \sqrt{(R - r)^2 + h^2}$

Sphere

- $Diameter = 2 \times Radius$
- $Surface\ Area = 4\pi r^2$
- $Volume = \frac{4}{3}\pi r^3$
- $Circumference\ of\ great\ circle = 2\pi r$

Hemisphere:

- The area of the flat circular base is = πr^2
- $Curved\ Surface\ Area = 2\pi r^2$
- $Surface\ Area = 2\pi r^2$
- $Volume = \frac{2}{3}\pi r^3$

Pyramid

Square-Based:

- $Base\ Area\ =\ s^2$ (where s is the side length of the square)
- $Slant\ Height(l)\ =\ \sqrt{(\frac{s}{2})^2 +\ h^2}$
- $Surface\ area\ =\ Base\ Area\ +\ Lateral\ Surface\ Area$
- $The\ total\ surface\ area\ =\ s^2 + 2 \times s\ \times l$
- $Volume\ = \frac{1}{3}\ \times s^2 \times h$

Triangular-Based (Tetrahedron):

- $Total\ surface\ Area\ = A_{base}\ +\ 3A_{lateral}$
- $For\ a\ regular\ triangular\ pyramid\ Total\ surface\ Area\ = a^2\sqrt{3}$

  where a is the side length of the equilateral triangle.

Prism

- $Lateral\ Surface\ Are\ = The\ sum\ of\ the\ areas\ of\ all\ the\ rectangular\ faces\ connecting\ bases$
- Surface Area = 2 × Base Area + Lateral Surface Area
- Volume = Base Area × Height